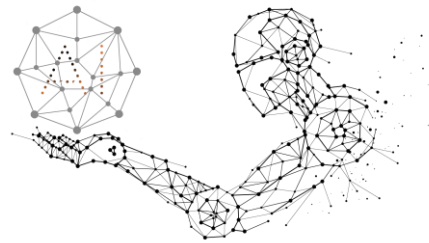
Applied Machine Learning

Course Introduction

دانشگاه تربیت مدرس
Tarbiat Modares University

Hossein Homaei
Department of Electrical & Computer Engineering



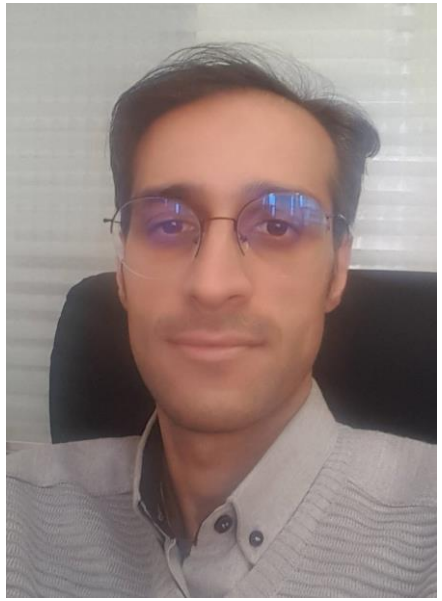


Welcome

Department of Electrical and Computer Engineering
Tarbiat Modares University



About me



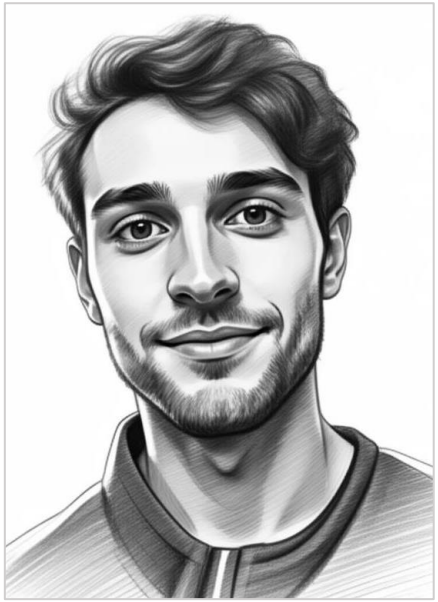
It’s me

Assistant professor in the Department of Electrical and Computer Engineering at Tarbiat Modares University since September 2020.

Feel free to contact me; It would be nice if I could help.

homaei@modares.ac.ir +98(21) 82883905

Room 6/55, 5th floor, End of the east corridor, Block 6



It’s fake me!
AI generated

About Assistants



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@saeedghff

About the Course

- Title
  - Applied Machine Learning

- Description
  - Introduction to the fundamental concepts and algorithms of classical machine learning
  - A combination of theory and practice
    - Programming language: Python



Goals

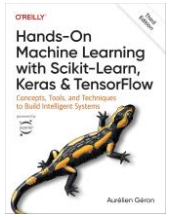
- By the end of this course, students will be able to:
  - Explain key machine learning concepts and terminology
  - Translate real-world challenges into machine learning problems
  - Build a full pipeline—from raw data to a deployed model
    - Select appropriate ML algorithms
    - Implement, train, and evaluate models using Python
    - Interpret the results

- Note: By mastering the application of classical algorithms, students build a solid framework for understanding more advanced topics like deep learning and reinforcement learning in the future.

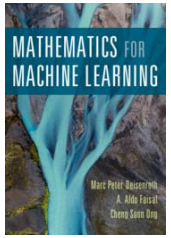


Main References

- Books



A. Geron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 3rd ed. O'Reilly Media, 2023.



M. P. Deisenroth, A. A. Faisal, and C. S. Ong, *Mathematics for machine learning*. Cambridge University Press, 2020.

...Main References

- Online
  - Machine Learning- Coursera Specialization
    - Instructors: Andrew Ng, Geoff Ladwig, Eddy Shyu, and Aarti Bagul
    - Stanford University and DeepLearning.AI



  - Mathematics for Machine Learning and Data Science- Coursera Specialization
    - Instructor: Luis Serrano
    - DeepLearning.AI



Grading Policy

Cheating = Zero point

- Exams
  - Midterm: 8 points
  - Final: 8 points
- Project: 4 points
  - In-person delivery
  - Evaluated very carefully and rigorously
    - All documents and codes are asked in detail
      - No answer = No grade
  - Late Policy:
    - Incur a 10% deduction for each day past the deadline
    - Projects submitted more than three days late will not be accepted unless accompanied by a valid, documented excuse (such as a medical emergency)



Homework?

- Yes, we have
  - But, No point!

- If you want to learn, you MUST do them.
  - Do it for yourselves, not to get points



教师打开门但你必须自己走进去

Student Expectations

- Active participation

- Collaborate with other students
  - Discuss concepts and general approaches with classmates
  - However, all submitted work (final project and exams) must be your own.
  - Plagiarism or any form of academic dishonesty, including copying code from another student, online sources, or chatbots, will result in a BIG zero.

- Spend at least 6 hours per week outside of class



Importance of the Course

- Machine learning is increasingly popular among
  - Academia
  - Industry
  - Daily life

- This course would provide the foundational skills required for roles such as Data Scientist, Machine Learning Engineer,....



Success stories

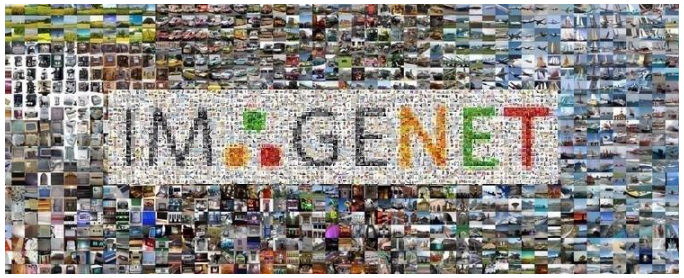
- Machine learning powers modern innovations in fields ranging from healthcare and finance to entertainment and science.

- Let’s review a few of them…

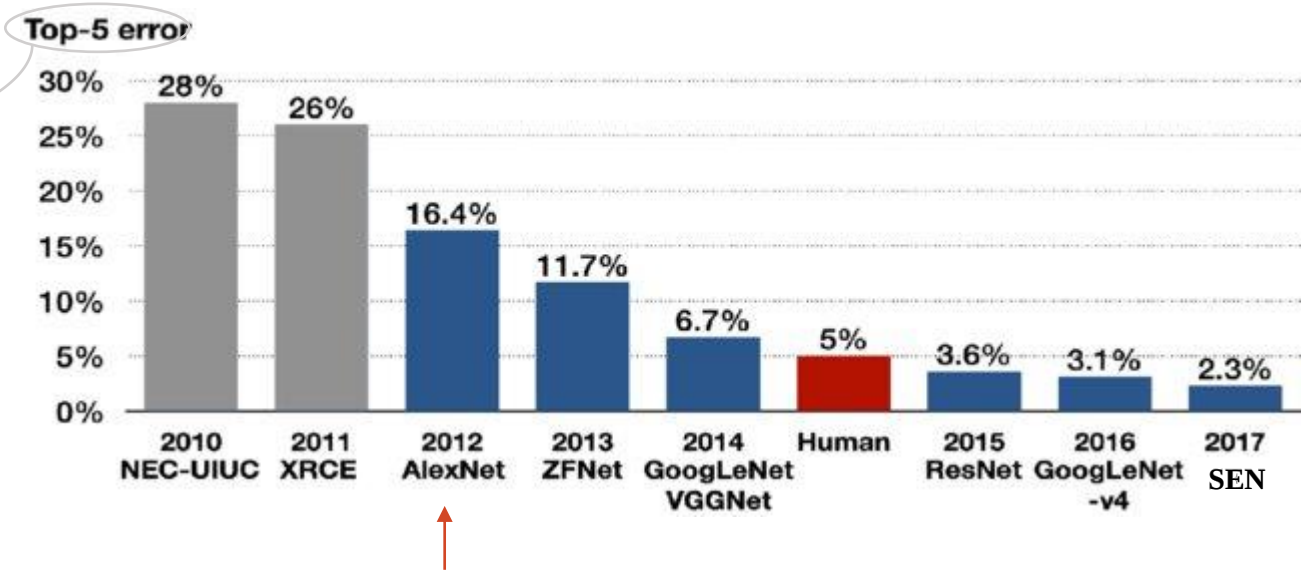


...Success stories

- ImageNet dataset
  - Goal: Developing computer vision models
  - Contains over 15 million high-resolution, labeled images
  - Large-Scale Visual Recognition Challenge
    - The image classification challenge
      - 1,000 object classes
      - 1,431,167 images

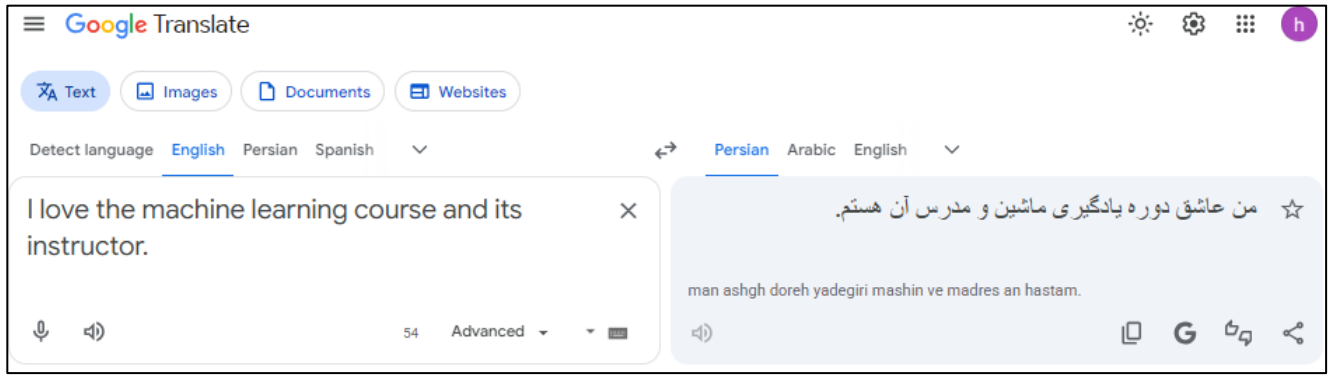


For each input image, the models output the 5 classes with the highest probabilities out of the 1,000 available classes; ultimately, in the error calculation, an image is considered an error if the object it contains is not among any of these 5 predicted classes.



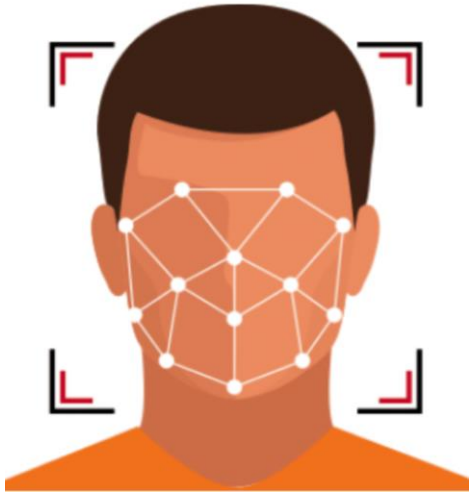
A Convolutional Neural Network (CNN)

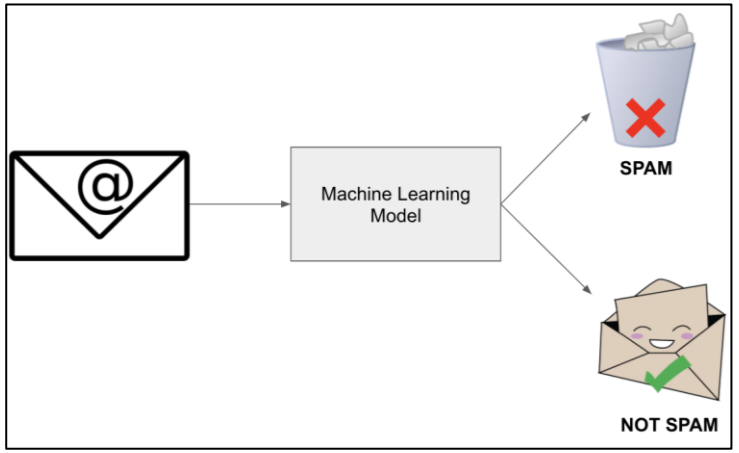
...Success stories





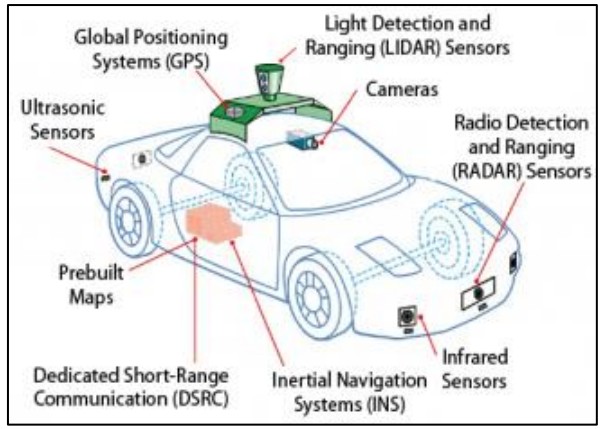




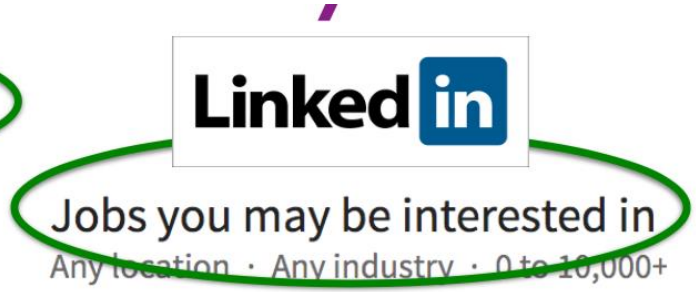


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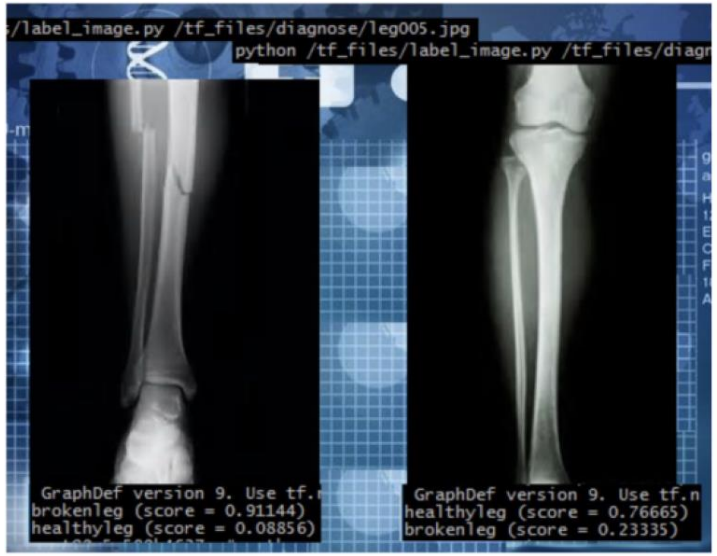






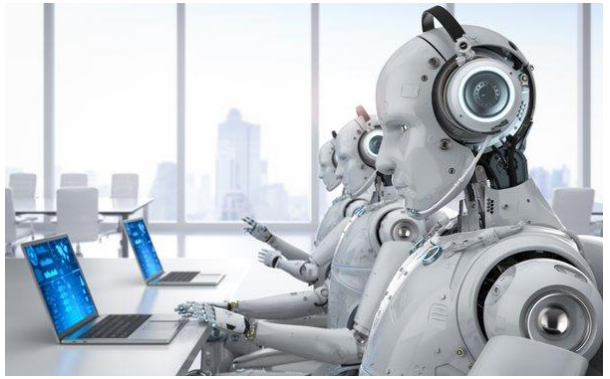


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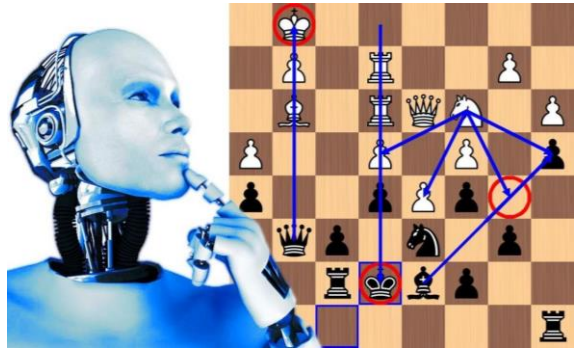






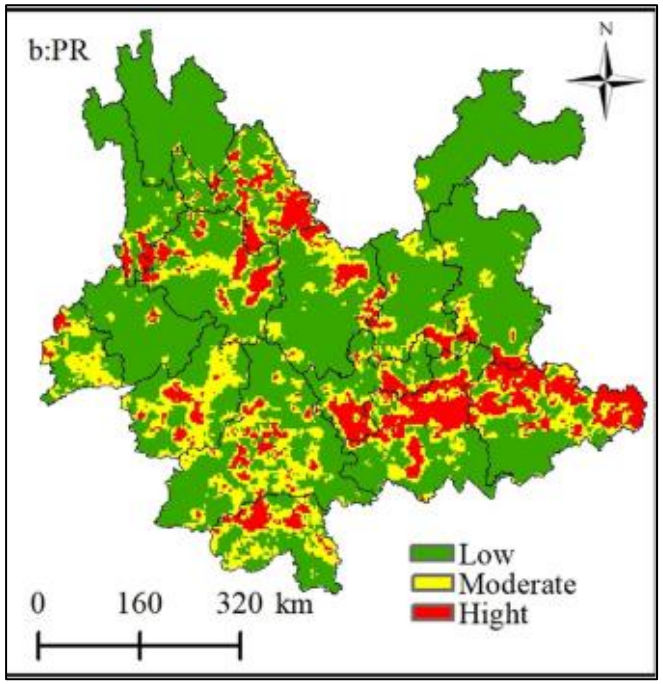


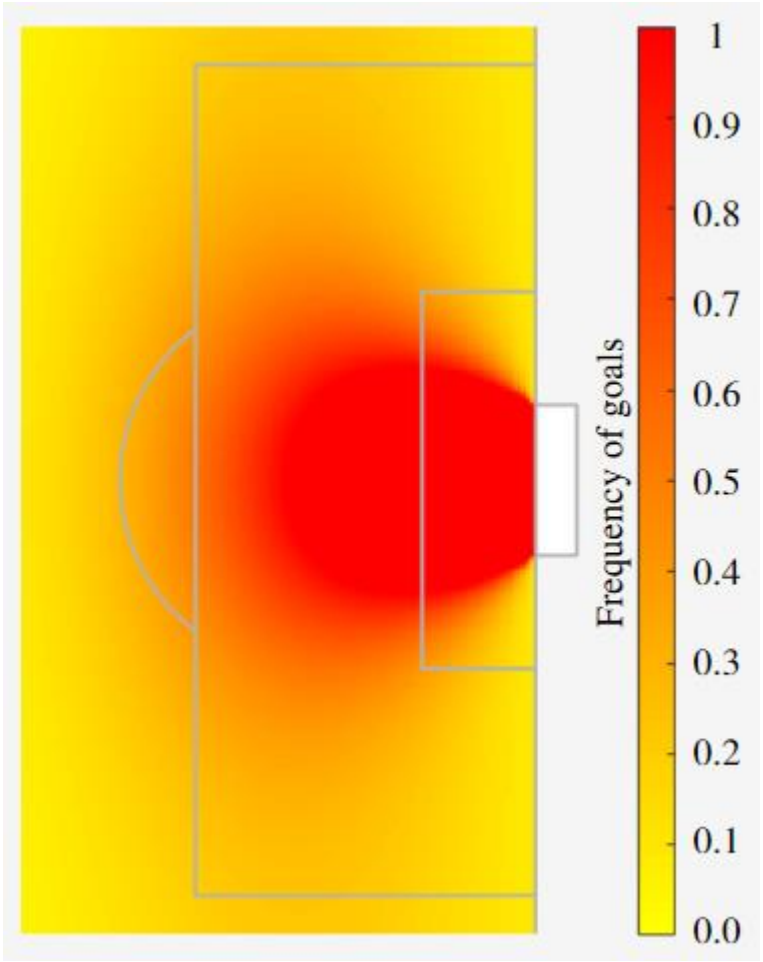




...Success stories







And Many More Applications…

- Article classification
- Flagging offensive comments on discussion forums
- Summarizing long documents
- Forecasting the company's revenue
- Fraud detection
- …

Ready to make this year amazing!

